Mehedi Ahmed

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SUMMARY

Senior Unity engineer with 8+ years shipping VR and AR apps on Meta Quest, Magic Leap, HoloLens, iOS, and Android, now co-founder and lead engineer of an independent game studio. Works mostly on the systems under the hood: NPC AI, rendering and shaders, networked multiplayer, and keeping standalone headsets at frame rate (18% FPS gain on Quest). Builds editor tools too, including a dialog editor that made designers 80% faster at authoring NPC behavior.

TECHNICAL SKILLS

Engines & Languages: Unity (through Unity 6), C#, HLSL, ShaderLab
Platforms: Meta Quest 2, Meta Quest 3, Meta Quest Pro (Meta XR SDK, OpenXR), PCVR (SteamVR, HTC Vive, Oculus Rift), Microsoft HoloLens, Magic Leap, Google Daydream, iOS (ARKit), Android (ARCore)
Unity Specializations: Meta XR SDK, OpenXR, XR Interaction Toolkit, AR Foundation, Hand Tracking, Passthrough and Mixed Reality, Spatial Anchors, In-Headset UI and HUDs, Event-Driven Architecture, Custom Editor Tooling, Addressable Asset System, Performance Profiling and Optimization, DOTS (ECS/Entities, Burst Compiler, Unity Job System), Animation Systems, Physics Simulation, Universal Render Pipeline (URP), HDRP, Shader Programming, Compute Shaders, GPU-Driven Rendering, Vulkan, OpenGL ES
Systems & AI: Hierarchical State Machines, Behavior Trees, NPC AI, Dialog Systems, Networked Multiplayer (Photon, Netcode for GameObjects), Real-Time Pose and Voice Streaming, VoIP Integration, REST API Integration, Multi-Device Calibration
Tools & Workflow: Git, Plastic SCM, Jira, Agile, Scrum, CI/CD, Claude Code, OpenAI Codex, Unity Test Runner, NUnit, Visual Studio, Rider, Xcode, Android Studio, Autodesk Maya, Adobe Photoshop
Familiar with: Unreal Engine 5, C++
Spoken Languages: English (Native), Bengali (Native)

WORK EXPERIENCE

Liminal Games Studio – Co-Founder & Lead Engineer *June 2025 – Present*

Two-person independent studio. Building Leave No Witnesses (PC stealth-action, in development) and Poly Blockade (iOS and Android tower defense, launching soon). liminalgamesstudio.com

- Built the NPC perception system in Burst-compiled C# on the Unity Job System, evaluating sight and sound for 100 agents inside a 1 ms frame budget
- Architected an event-driven, assembly-isolated codebase where AI, simulation, and rendering communicate only through shared interfaces and primitive ids; a machine-checked boundary test suite fails any cross-system reference in a single Test Runner pass
- Designed a per-actor physiological state model (pain, blood loss, vitality, per-zone function) that replaces scalar health, so outcomes depend on weapon, body zone, and structure hit; lazy tick registration keeps uninjured actors at zero per-frame cost
- Designed a semantic octree for spatial queries so AI can reason about cover, sightlines, and evidence without traversing the scene graph every frame
- Built a custom URP render pipeline with fully GPU-driven compute passes and indirect draw args, so no per-frame counts return to the CPU

SimX – Unity Engineer *August 2023 – October 2025*

- Shipped enterprise VR medical training applications on Meta Quest in Unity (C#) using the Meta XR SDK, owning features from design through production deployment and integrating with backend services for content and session management
- Built custom Unity Editor tooling (C#, JSON, Addressables) for dynamic patient equipment management, letting the team configure asset loading 60% faster than manual workflows
- Led performance optimization on Quest using Unity Profiler and custom tooling, delivering an 18% FPS improvement through GC reduction, memory management, material and texture sharing, and draw call reduction

- Engineered a proprietary event-driven animation system (C#) for VR, replacing Unity's Animator to cut runtime cost and support application-specific state and blending behavior
- Established company-wide AI development standards and standardized context files adopted across the engineering organization
- Developed a texture combiner (C#, shaders) that procedurally generates wound textures on patient models, removing per-variant skinned mesh work and accelerating content development by 40%
- Accelerated development velocity by 30% through process improvements, custom tooling, and pipeline enhancements while maintaining code quality standards

WIN Reality – Senior Unity Engineer *May 2022 – December 2022*

- Developed VR training features in Unity (C#) for Meta Quest 2 on a consumer sports training application with thousands of active users
- Developed custom URP shaders (HLSL, ShaderLab) for dynamic visual effects, including a performance-optimized trail renderer for real-time ball tracking
- Rebuilt the in-headset UI around VR interaction patterns (world-space menus, interactables, and HUD elements), improving user engagement metrics by 13%
- Maintained code quality across the engineering team through code reviews (Plastic SCM), mentorship, and shared engineering frameworks for high-performance modules

Avrio Analytics – Senior AR Engineer *May 2021 – May 2022*

- Delivered government-facing AR training applications for Magic Leap, owning project planning, Unity development (C#), and stakeholder communication
- Architected the real-time networking (Photon) for colocated multiplayer AR, streaming pose and VoIP audio between headsets, and built the calibration flow that brings multiple devices onto a shared coordinate frame via fiducial marker scanning
- Designed state-driven NPC behavior (hierarchical behavior trees, C#, custom framework) for law enforcement de-escalation training with multi-stage adaptive interactions
- Led a cross-functional team of 4 engineers using Agile/Scrum to release multiple AR training applications on schedule
- Developed a node-based dialog tree editor (C#, Unity Editor) controlling AI dialog and animations for training scenarios, letting designers build complex behaviors 80% faster

Independent – Contract VR/AR Developer *December 2019 – April 2021*

Delivered VR and AR applications for enterprise clients across healthcare and medical training domains.

Tipping Point Media: Developed iOS AR applications (Unity, ARKit, C#) for enterprise clients, including custom fluid simulation shaders (HLSL, ShaderLab) for liquid medication visualization in AR training.

Pixacore: Built a VR training application (Unity, C#) for Oculus Quest teaching healthcare professionals CAR T-Cell cancer therapy procedures.

Code & Craft, Inc – VR/AR Developer *January 2017 – August 2018*

- Delivered multiple consumer-facing Unity (C#) applications from prototype through production release, managing technical scope and quality
- Launched one of the first commercial iOS AR applications built on Apple ARKit, a retail experience released on the App Store

SELECTED PROJECTS

Operation Viewpoint – Asymmetrical multiplayer VR stealth game (HTC Vive, SteamVR, Unity, C#). Lead programmer; built the AI systems and multiplayer networking and contributed to core game design.

The Hollows – VR cave exploration experience (HTC Vive, SteamVR, Unity, C#) with custom locomotion mechanics and optimized real-time lighting.

The Virtual Memory Palace – Mobile VR spatial cognition training tool (Google Daydream, Unity, C#) built at MIT Media Lab 'Reality Virtually' Hackathon.

EDUCATION

University of Massachusetts Amherst – Bachelor of Science in Informatics
Multidisciplinary program combining computer science, mathematics, studio art, and art history